TCP/IP Suite Theory

1.Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.

ANS:

Wireshark Packet Capture – Flipkart Homepage

Aim: To capture network traffic while refreshing the Flipkart homepage and identify packets from four TCP/IP model layers.

Procedure:

1. Open Wireshark and select your active Wi-Fi or Ethernet interface.
2. Click Start Capture.
3. Open your browser, visit https://www.flipkart.com, and refresh the homepage.
4. Return to Wireshark and click Stop Capture.
5. Examine the Protocol and Info columns and expand the packet headers.

Packet Identification Table

TCP/IP Layer	Example Protocol	Header Details to Check
Application	DNS	Query name, such as www.flipkart.com
Transport	TCP	Source port, destination port, sequence number
Internet	IP (IPv4/IPv6)	Source IP address, destination IP address
Network Access	Ethernet or Wi-Fi	Source and destination MAC addresses, if visible

Useful Wireshark display filters:

- DNS: dns
- TCP: tcp
- IP: ip
- Ethernet: eth

2.Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).

ANS:

Comparison of OSI and TCP/IP Models

OSI Model Layer	TCP/IP Model Layer	Real-World Example
7. Application	Application	WhatsApp — sending a message
6. Presentation	Application	WhatsApp — message encryption and data formatting
5. Session	Application	WhatsApp — managing a video-call session
4. Transport	Transport	WhatsApp — delivering data using TCP or UDP
3. Network	Internet	Zomato — routing order data using IP addresses
2. Data Link	Network Access	Laptop — sending frames over Wi-Fi
1. Physical	Network Access	Phone — transmitting data through Wi-Fi radio signals

**3.Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines.

Hint: Look for fields like Source Port, Destination Port, Sequence Number, etc.**

ANS:

TCP Segment Header – Transport Layer

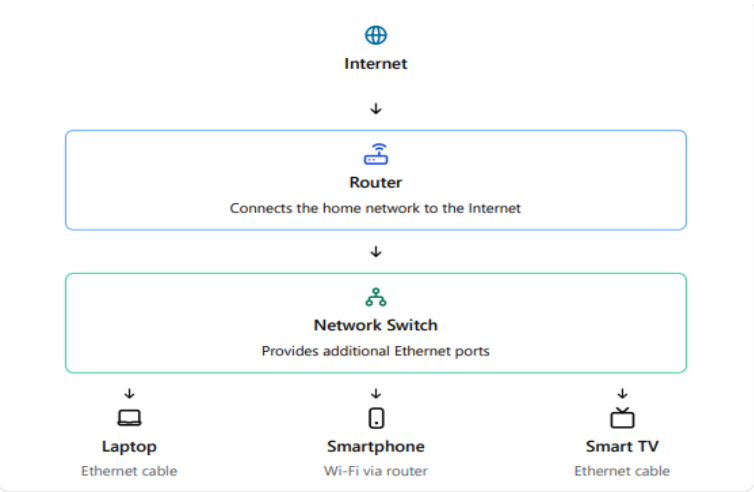
TCP Header Field	Function
Source Port	Identifies the port number of the sending application.
Destination Port	Identifies the port number of the receiving application.
Sequence Number	Indicates the position of data bytes in the TCP stream and helps arrange data in order.
Acknowledgment Number	Indicates the next byte expected from the sender.
Header Length (Data Offset)	Specifies the size of the TCP header.
Flags (Control Bits)	Control connection operations, such as SYN to start and FIN to close a connection.
Window Size	Indicates how much data the receiver can accept before an acknowledgment is needed.
Checksum	Detects errors in the TCP header and data.

Urgent Pointer	Indicates the end of urgent data when the URG flag is set.
Options	Carries optional settings, such as Maximum Segment Size (MSS).
Data (Payload)	Contains the application data carried by the TCP segment; it is not part of the TCP header itself.

4.Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).

ANS:

TCP/IP Model – Sending a Message to Instagram



5.Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.

ANS:

Wireshark Packet Analysis

Selected Packet: TCP Packet

- TCP/IP Layer: Transport Layer
- How I Identified It: I checked the Protocol column in Wireshark, where the packet was labelled TCP. Its header contains source port, destination port, and sequence number fields.
- Real-World Action: When I send a message on WhatsApp, TCP may carry data between the device and a server, depending on the connection used.